

Lecture Notes: Differential Equations

1. Introduction to Differential Equations

Differential equations are equations that relate a function to its derivatives. They are fundamental in the mathematical modeling of physical, biological, and economic phenomena, among others.

Classification of Differential Equations

1. **Ordinary Differential Equations (ODEs):** involve a function of a single independent variable and its derivatives.
2. **Partial Differential Equations (PDEs):** contain partial derivatives of a function with multiple independent variables.

Order and Linearity

- **Order:** defined by the highest derivative present in the equation.
 - **Linear vs. Nonlinear:** An equation is linear if the function and its derivatives appear in a linear manner (without products or nonlinear functions of them).
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2. First-Order Ordinary Differential Equations

These are equations of the form:

$$F(x, y, y') = 0$$

or, in explicit form,

$$y' = f(x, y)$$

Types of First-Order Equations

1. **Separable Equations:** can be rewritten as

$$g(y)dy = f(x)dx$$

and solved by direct integration.

2. **Homogeneous Equations:** of the form

$$y' = f(y/x)$$

where the substitution $v = y/x$ is used.

3. First-Order Linear Equations:

$$y' + P(x)y = Q(x)$$

which are solved using an integrating factor.

4. Exact Equations: satisfy

$$M(x, y)dx + N(x, y)dy = 0$$

and the exactness condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

3. Solution Methods and Applications

Solution Methods

1. **Direct integration** (for separable equations).
2. **Using integrating factors** (for linear equations).
3. **Applying appropriate variable substitutions** (for homogeneous or reducible equations).
4. **Checking exactness conditions and finding potential functions** (for exact equations).

Common Applications

- **Exponential Growth and Decay:**

$$\frac{dy}{dt} = ky$$

with solution $y(t) = y_0 e^{kt}$.

- **Newton's Law of Cooling:**

$$\frac{dT}{dt} = -k(T - T_\infty)$$

models the temperature change of a body.

- **Falling Bodies with Air Resistance:**

$$m \frac{dv}{dt} = mg - kv$$

describes the velocity of a falling object.

This topic lays the foundation for higher-order equations and partial differential equations.